

12.1 Performance Requirements

IEEE 830 Reference: Section 3.3 - Non-Functional Requirements: Performance **Source Documents:** SUPP-039 (Infrastructure, Performance & Security Requirements) **Version:** 1.0 **Last Updated:** 2026-01-01

12.1.1 Purpose

This section defines measurable performance requirements for NewPOPSys v1, including response time targets, throughput limits, and database query performance expectations. All requirements include specific targets and verification methods.

12.1.2 API Response Time Requirements

12.1.2.1 Response Time SLAs by Endpoint Category

NFR-ID	Endpoint Category	p50 Target	p95 Target	p99 Target	Max Allowed	Priority
NFR-PERF-001	Simple reads (GET list endpoints)	< 50ms	< 150ms	< 300ms	1s	Must Have
NFR-PERF-002	Complex reads (dashboard aggregations)	< 100ms	< 300ms	< 500ms	2s	Must Have
NFR-PERF-003	Write operations (create/update)	< 100ms	< 250ms	< 500ms	2s	Must Have
NFR-PERF-004	Photo upload initiation (presigned URL)	< 200ms	< 500ms	< 1s	3s	Must Have
NFR-PERF-005	Report generation (synchronous)	< 2s	< 5s	< 10s	30s	Should Have
NFR-PERF-006	Export generation (asynchronous)	N/A	N/A	N/A	5 min	Should Have

12.1.2.2 Verification Methods

NFR-ID	Verification Method
NFR-PERF-001 through NFR-004	Automated load testing with k6/Artillery in CI/CD pipeline
NFR-PERF-005	Integration test suite with timing assertions
NFR-PERF-006	Background job monitoring via BullMQ dashboard metrics

12.1.3 Throughput Requirements

12.1.3.1 System Throughput Targets

NFR-ID	Metric	Normal Target	Burst Capacity	Priority
NFR-PERF-010	API requests per second	100 rps	500 rps	Must Have
NFR-PERF-011	Concurrent WebSocket connections	200	500	Should Have
NFR-PERF-012	Photo uploads per minute	50	200	Must Have
NFR-PERF-013	Database queries per second	500 qps	2,000 qps	Must Have
NFR-PERF-014	Background jobs per minute	100	500	Should Have

12.1.3.2 Concurrent User Capacity

NFR-ID	Store Count	Concurrent Users	API Instances	Priority
NFR-PERF-015	100 stores	10 users	2 instances	Must Have (Pilot)
NFR-PERF-016	500 stores	50 users	2-4 instances	Must Have (v1)
NFR-PERF-017	2,000 stores	200 users	4-8 instances	Should Have (v1.1)
NFR-PERF-018	10,000 stores	1,000 users	8-16 instances	Nice to Have (v2)

12.1.3.3 Verification Methods

NFR-ID	Verification Method
NFR-PERF-010 through NFR-014	Load testing with realistic traffic patterns
NFR-PERF-015 through NFR-018	Staged pilot with progressive user onboarding

12.1.4 Database Performance Requirements

12.1.4.1 Query Performance Targets

NFR-ID	Query Type	Target Time	Index Strategy	Priority
NFR-PERF-020	Primary key lookup	< 1ms	B-tree on UUID PKs	Must Have
NFR-PERF-021	Campaign list (filtered by tenant)	< 10ms	Composite on tenant_id + status	Must Have
NFR-PERF-022	Store search by name/number	< 20ms	GIN index on name, store_number	Must Have
NFR-PERF-023	Rollup aggregations (dashboards)	< 100ms	Materialized views with refresh	Should Have
NFR-PERF-024	Full-text search	< 50ms	pg_trgm + GIN index	Should Have

12.1.4.2 Connection Pool Configuration

NFR-ID	Parameter	Value	Notes	Priority
NFR-PERF-025	Min connections per instance	5	Warm pool for immediate use	Must Have
NFR-PERF-026	Max connections per instance	20	Prevent connection exhaustion	Must Have
NFR-PERF-027	Connection timeout	5s	Fail fast on unavailability	Must Have
NFR-PERF-028	Query timeout (default)	30s	Prevent runaway queries	Must Have

12.1.4.3 Verification Methods

- Query performance monitoring via `pg_stat_statements` extension
 - Slow query logging for queries exceeding 100ms
 - Explain plan analysis for all list/aggregation endpoints
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12.1.5 Photo Processing Performance

12.1.5.1 Photo Pipeline Targets

NFR-ID	Operation	Target Time	Implementation Strategy	Priority
NFR-PERF-030	Upload to S3 (10MB photo)	< 5s	Direct-to-S3 presigned URL	Must Have
NFR-PERF-031	Thumbnail generation	< 2s	Lambda trigger on S3 upload	Must Have
NFR-PERF-032	EXIF metadata extraction	< 500ms	Lambda in upload pipeline	Should Have
NFR-PERF-033	CDN cache hit response	< 50ms	CloudFront edge caching	Must Have

12.1.5.2 Photo Size Limits

NFR-ID	Constraint	Value	Priority
NFR-PERF-034	Maximum single photo size	15 MB	Must Have
NFR-PERF-035	Photos per survey submission	20 max	Must Have
NFR-PERF-036	Total upload per session	100 MB	Should Have

12.1.5.3 Verification Methods

- S3 upload timing instrumentation with OpenTelemetry
 - Lambda CloudWatch metrics for processing duration
 - CDN cache hit ratio monitoring via CloudFront analytics
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12.1.6 Page Load Performance

12.1.6.1 Web Vitals Targets

NFR-ID	Metric	Target	Priority
NFR-PERF-040	Largest Contentful Paint (LCP)	< 2.5s	Must Have
NFR-PERF-041	First Input Delay (FID)	< 100ms	Must Have
NFR-PERF-042	Cumulative Layout Shift (CLS)	< 0.1	Should Have
NFR-PERF-043	Time to First Byte (TTFB)	< 600ms	Must Have
NFR-PERF-044	First Contentful Paint (FCP)	< 1.8s	Should Have

12.1.6.2 PWA Performance (Mobile Store App)

NFR-ID	Metric	Target	Priority
NFR-PERF-045	Initial app load (cold start)	< 3s on 4G	Must Have
NFR-PERF-046	Cached app load (warm start)	< 1s	Must Have
NFR-PERF-047	Survey form render time	< 500ms	Must Have
NFR-PERF-048	Offline data sync (when reconnecting)	< 10s for 50 entries	Should Have

12.1.6.3 Verification Methods

- Lighthouse CI integration in deployment pipeline
 - Real User Monitoring (RUM) via Sentry Performance
 - Synthetic monitoring for critical user journeys
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12.1.7 Caching Strategy

12.1.7.1 Cache Layer Requirements

NFR-ID	Cache Layer	TTL	Invalidation Strategy	Priority
NFR-PERF-050	CDN (static assets)	1 year (immutable)	Content-hash filenames	Must Have
NFR-PERF-051	CDN (API responses)	5 min	Cache-Control headers	Should Have
NFR-PERF-052	Redis (session data)	15 min sliding	Explicit invalidation on logout	Must Have
NFR-PERF-053	Redis (list data)	1 min	Invalidate on write	Should Have
NFR-PERF-054	Application (reference data)	5 min	Background refresh	Should Have

12.1.7.2 Cache Hit Ratio Targets

NFR-ID	Cache Layer	Target Hit Ratio	Priority
NFR-PERF-055	CDN static assets	> 95%	Must Have
NFR-PERF-056	Redis session cache	> 99%	Must Have
NFR-PERF-057	API response cache	> 60%	Should Have

12.1.8 Background Job Performance

12.1.8.1 Job Processing Targets

NFR-ID	Job Type	Target Duration	Queue	Priority
NFR-PERF-060	Email notification send	< 2s	default	Must Have
NFR-PERF-061	Webhook delivery attempt	< 5s	default	Must Have
NFR-PERF-062	Thumbnail generation	< 10s	default	Must Have
NFR-PERF-063	CSV export (1000 rows)	< 30s	long	Should Have
NFR-PERF-064	Full export bundle	< 5 min	long	Should Have
NFR-PERF-065	Media purge job (per campaign)	< 2 min	long	Should Have

12.1.8.2 Queue Depth Limits

NFR-ID	Queue	Max Depth (Alert)	Max Depth (Critical)	Priority
NFR-PERF-066	Default queue	1,000 jobs	5,000 jobs	Must Have
NFR-PERF-067	Long-running queue	100 jobs	500 jobs	Should Have
NFR-PERF-068	Dead-letter queue	10 jobs	50 jobs	Must Have

12.1.9 Monitoring and Alerting

12.1.9.1 Performance Monitoring Requirements

NFR-ID	Requirement	Implementation	Priority
NFR-PERF-070	Request tracing	OpenTelemetry with request ID correlation	Must Have
NFR-PERF-071	Performance dashboards	CloudWatch or Grafana with key metrics	Must Have
NFR-PERF-072	Anomaly detection	Automated alerts for p95 degradation	Should Have
NFR-PERF-073	Real-time error tracking	Sentry integration with performance context	Must Have

12.1.9.2 Alert Thresholds

NFR-ID	Metric	Warning Threshold	Critical Threshold	Priority
NFR-PERF-074	API p95 response time	> 500ms	> 2s	Must Have
NFR-PERF-075	Error rate	> 1%	> 5%	Must Have
NFR-PERF-076	Database connection pool usage	> 70%	> 90%	Must Have
NFR-PERF-077	CPU utilization (container)	> 70%	> 90%	Must Have
NFR-PERF-078	Memory utilization (container)	> 80%	> 95%	Must Have

12.1.10 Cross-References

Reference	Description
SUPP-039	Authoritative infrastructure and performance requirements
SUPP-012	Technology stack selections
Section 3.3	Technology Stack documentation
Section 12.4	Scalability requirements
Section 12.5	Availability requirements

Document Status: Complete IEEE 830 Compliance: Section 3.3 - Performance Requirements

12.2 Security Requirements

IEEE 830 Reference: Section 3.3 - Non-Functional Requirements: Security **Source Documents:** SUPP-039 (Infrastructure, Performance & Security Requirements), SUPP-003 (RBAC) **Version:** 1.0 **Last Updated:** 2026-01-01

12.2.1 Purpose

This section defines security requirements for NewPOPSys v1, covering authentication, authorization, data protection, compliance, and security operations. All requirements include specific implementation targets and verification methods.

12.2.2 Authentication Requirements

12.2.2.1 Primary Authentication

NFR-ID	Requirement	Specification	Priority
NFR-SEC-001	Web application authentication	Server-side session cookies	Must Have
NFR-SEC-002	API authentication	JWT tokens with RS256 signing	Must Have
NFR-SEC-003	Access token lifetime	15 minutes	Must Have
NFR-SEC-004	Refresh token lifetime	7 days	Must Have
NFR-SEC-005	Session storage	Redis-backed with sliding expiration	Must Have

12.2.2.2 Multi-Factor Authentication

NFR-ID	Requirement	Specification	Priority
NFR-SEC-010	MFA for admin roles	TOTP required for PSP Admin, Platform Admin, Brand Admin	Must Have
NFR-SEC-011	MFA for store users	Optional, user-configurable	Should Have
NFR-SEC-012	MFA recovery	Backup codes (10 single-use codes)	Must Have
NFR-SEC-013	MFA enrollment grace period	7 days for new admin accounts	Should Have

12.2.2.3 Password Policy

NFR-ID	Requirement	Specification	Priority
NFR-SEC-020	Minimum password length	12 characters	Must Have
NFR-SEC-021	Password complexity	At least 1 uppercase, 1 lowercase, 1 number	Must Have
NFR-SEC-022	Password history	Prevent reuse of last 5 passwords	Should Have
NFR-SEC-023	Password hashing	Argon2id or bcrypt (cost factor 12)	Must Have
NFR-SEC-024	Compromised password check	Integration with HaveIBeenPwned API	Nice to Have

12.2.2.4 Account Lockout

NFR-ID	Requirement	Specification	Priority
NFR-SEC-025	Failed login threshold	5 consecutive failures	Must Have
NFR-SEC-026	Lockout duration	15 minutes (progressive)	Must Have

NFR-SEC-027	Lockout notification	Email notification to user	Should Have
NFR-SEC-028	Admin unlock capability	Manual unlock via admin console	Must Have

12.2.2.5 Verification Methods

- Automated security testing for auth flows in CI/CD
 - Penetration testing before launch
 - Session hijacking prevention verification
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12.2.3 Authorization Requirements

12.2.3.1 Role-Based Access Control (RBAC)

NFR-ID	Requirement	Specification	Priority
NFR-SEC-030	Permission enforcement	Server-side RBAC at API layer	Must Have
NFR-SEC-031	Role hierarchy	Defined role inheritance model	Must Have
NFR-SEC-032	Permission granularity	Resource + action level (e.g., campaign:read)	Must Have
NFR-SEC-033	Dynamic permission loading	No hardcoded permissions in UI	Must Have

12.2.3.2 Tenant Isolation

NFR-ID	Requirement	Specification	Priority
NFR-SEC-035	Data isolation	Tenant ID enforced on all queries	Must Have
NFR-SEC-036	Cross-tenant prevention	Middleware validates tenant context	Must Have
NFR-SEC-037	Tenant-scoped URLs	All resources include tenant context	Must Have
NFR-SEC-038	Audit of cross-tenant attempts	Log and alert on violations	Must Have

12.2.3.3 API Authorization

NFR-ID	Requirement	Specification	Priority
NFR-SEC-040	Integration API keys	Hashed storage, scoped permissions	Must Have
NFR-SEC-041	API key rotation	Support for key rotation without downtime	Should Have
NFR-SEC-042	Webhook signatures	HMAC-SHA256 with shared secret	Must Have
NFR-SEC-043	Idempotency keys	Required for webhook processing	Must Have

12.2.3.4 Verification Methods

- Authorization bypass testing (horizontal and vertical)
 - Automated RBAC matrix validation
 - Tenant isolation penetration testing
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12.2.4 Data Protection Requirements

12.2.4.1 Encryption at Rest

NFR-ID	Data Type	Encryption Method	Priority
NFR-SEC-050	Database (RDS PostgreSQL)	AES-256 via RDS encryption	Must Have
NFR-SEC-051	Object storage (S3)	SSE-S3 or SSE-KMS	Must Have
NFR-SEC-052	Secrets and credentials	AWS KMS envelope encryption	Must Have
NFR-SEC-053	Database backups	Encrypted snapshots	Must Have
NFR-SEC-054	Application logs	CloudWatch encryption enabled	Must Have

12.2.4.2 Encryption in Transit

NFR-ID	Requirement	Specification	Priority
NFR-SEC-055	HTTPS enforcement	TLS 1.3 minimum, HTTP redirects to HTTPS	Must Have
NFR-SEC-056	Database connections	TLS required for all connections	Must Have
NFR-SEC-057	Internal service communication	mTLS or signed JWTs	Should Have
NFR-SEC-058	S3 access	HTTPS-only bucket policy	Must Have

12.2.4.3 Sensitive Data Handling

NFR-ID	Requirement	Specification	Priority
NFR-SEC-060	PII field identification	Annotated in data model	Must Have
NFR-SEC-061	PII in logs	Automatically redacted	Must Have
NFR-SEC-062	PII in error messages	Never exposed in API responses	Must Have
NFR-SEC-063	Sensitive data in URLs	Never include passwords or tokens	Must Have

12.2.4.4 Verification Methods

- TLS certificate validation in monitoring
- Encryption verification via AWS Config rules
- Log audit for PII leakage

12.2.5 OWASP Top 10 Mitigation

12.2.5.1 Security Controls Matrix

NFR-ID	OWASP Risk	Mitigation Strategy	Priority
NFR-SEC-070	A01: Broken Access Control	RBAC at API layer, tenant isolation, authorization middleware	Must Have
NFR-SEC-071	A02: Cryptographic Failures	TLS 1.3, AES-256 at rest, no sensitive data in URLs	Must Have
NFR-SEC-072	A03: Injection	Parameterized queries via Drizzle ORM, input validation	Must Have
NFR-SEC-073	A04: Insecure Design	Threat modeling, secure defaults, principle of least privilege	Must Have
NFR-SEC-074	A05: Security Misconfiguration	Infrastructure as Code, security scanning in CI/CD	Must Have
NFR-SEC-075	A06: Vulnerable Components	Dependabot alerts, regular patching, SCA scanning	Must Have
NFR-SEC-076	A07: Auth Failures	Rate limiting, account lockout, strong password policy	Must Have
NFR-SEC-077	A08: Data Integrity Failures	Signed deployments, audit logging, HMAC webhooks	Must Have
NFR-SEC-078	A09: Logging Failures	Centralized logging, anomaly detection, log integrity	Must Have
NFR-SEC-079	A10: SSRF	URL validation, egress filtering, allowlists	Must Have

12.2.5.2 Verification Methods

- OWASP ZAP automated scanning in CI/CD
- Manual penetration testing quarterly
- Dependency vulnerability scanning with Snyk/Dependabot

12.2.6 Audit Logging Requirements

12.2.6.1 Audit Event Categories

NFR-ID	Event Category	Events Captured	Priority
NFR-SEC-080	Authentication	Login, logout, failed attempts, MFA events	Must Have
NFR-SEC-081	Authorization	Permission denied, role changes, access escalation	Must Have
NFR-SEC-082	Data access	Read of sensitive resources, exports, bulk operations	Must Have

NFR-SEC-083	Data modification	Create, update, delete of business entities	Must Have
NFR-SEC-084	Administrative	User management, configuration changes, impersonation	Must Have
NFR-SEC-085	System events	Backup, purge, integration sync	Should Have

12.2.6.2 Audit Log Fields

NFR-ID	Requirement	Specification	Priority
NFR-SEC-086	Timestamp	UTC with millisecond precision	Must Have
NFR-SEC-087	Actor identification	User ID, role, IP address, user agent	Must Have
NFR-SEC-088	Action details	Resource type, resource ID, action type	Must Have
NFR-SEC-089	Request correlation	Request ID for distributed tracing	Must Have
NFR-SEC-090	Tenant context	Tenant ID for multi-tenant isolation	Must Have

12.2.6.3 Audit Log Protection

NFR-ID	Requirement	Specification	Priority
NFR-SEC-091	Log immutability	Append-only, no deletion capability	Must Have
NFR-SEC-092	Log retention	2 years minimum	Must Have
NFR-SEC-093	Log access control	Separate from application permissions	Must Have
NFR-SEC-094	Log integrity	Tamper detection via checksums	Should Have

12.2.7 Security Operations

12.2.7.1 Vulnerability Management

NFR-ID	Requirement	Specification	Priority
NFR-SEC-100	Dependency scanning	Automated in CI/CD via Dependabot/Snyk	Must Have
NFR-SEC-101	Container scanning	Image vulnerability scanning before deployment	Must Have
NFR-SEC-102	Critical vulnerability SLA	Patch within 24 hours	Must Have
NFR-SEC-103	High vulnerability SLA	Patch within 7 days	Must Have

NFR-SEC-104	Medium vulnerability SLA	Patch within 30 days	Should Have
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12.2.7.2 Penetration Testing

NFR-ID	Testing Phase	Timing	Scope	Priority
NFR-SEC-105	Pre-launch	Before beta deployment	Full application + infrastructure	Must Have
NFR-SEC-106	Quarterly	Every 3 months	Critical paths + new features	Should Have
NFR-SEC-107	Annual	Yearly	Full scope + social engineering	Nice to Have

12.2.7.3 Security Monitoring

NFR-ID	Requirement	Specification	Priority
NFR-SEC-110	WAF deployment	AWS WAF with OWASP rule set	Must Have
NFR-SEC-111	Rate limiting	Per-endpoint and per-IP limits	Must Have
NFR-SEC-112	Anomaly detection	Automated alerts for unusual patterns	Should Have
NFR-SEC-113	Failed auth monitoring	Alert on brute force patterns	Must Have

12.2.8 Compliance Requirements

12.2.8.1 Regulatory Compliance

NFR-ID	Standard	Applicability	v1 Status	Priority
NFR-SEC-120	GDPR	EU store/customer data	Basic compliance	Must Have
NFR-SEC-121	CCPA	California stores	Basic compliance	Must Have
NFR-SEC-122	SOC 2 Type II	Enterprise customers	Future (v2+)	Nice to Have
NFR-SEC-123	PCI DSS	N/A (no payment processing)	Not applicable	N/A

12.2.8.2 GDPR Specific Requirements

NFR-ID	Requirement	Implementation	Priority
NFR-SEC-125	Right to access	Data export functionality	Must Have
NFR-SEC-126	Right to deletion	Data purge workflow	Must Have
NFR-SEC-127	Data portability	Standard export formats (CSV, JSON)	Should Have
NFR-SEC-128	Consent management	Consent tracking for communications	Should Have

NFR-SEC-129	Privacy by design	Data minimization, purpose limitation	Must Have
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12.2.8.3 Verification Methods

- Compliance checklist review
- Third-party audit (for SOC 2)
- Privacy impact assessment

12.2.9 Incident Response

12.2.9.1 Security Incident Classification

NFR-ID	Severity	Description	Response Time	Priority
NFR-SEC-130	Critical	Active breach, data exfiltration	Immediate (< 1 hour)	Must Have
NFR-SEC-131	High	Vulnerability exploitation attempt	< 4 hours	Must Have
NFR-SEC-132	Medium	Suspicious activity, policy violation	< 24 hours	Should Have
NFR-SEC-133	Low	Security misconfiguration, audit finding	< 7 days	Should Have

12.2.9.2 Incident Response Requirements

NFR-ID	Requirement	Specification	Priority
NFR-SEC-135	Incident runbook	Documented response procedures	Must Have
NFR-SEC-136	Escalation path	Defined escalation contacts and thresholds	Must Have
NFR-SEC-137	Breach notification	Process for regulatory notification (GDPR 72hr)	Must Have
NFR-SEC-138	Post-incident review	Blameless retrospective within 7 days	Should Have

12.2.10 Cross-References

Reference	Description
SUPP-039	Infrastructure and security requirements source
SUPP-003	RBAC and permissions matrix
SUPP-031	Supportability policies and impersonation
Section 4.2	Permission Matrix documentation
Section 4.3	Authentication Flows documentation
Section 12.6	Data Retention requirements

12.3 Accessibility Requirements

IEEE 830 Reference: Section 3.3 - Non-Functional Requirements: Accessibility **Source Documents:** SUPP-038 (Section 508 Accessibility Scaffold) **Version:** 1.0 **Last Updated:** 2026-01-01

12.3.1 Purpose

This section defines accessibility requirements for NewPOPSys v1, ensuring the platform is usable by people with disabilities and compliant with federal accessibility mandates. The v1 release establishes foundational patterns enabling full WCAG 2.1 Level AA compliance in v1.1.

12.3.2 Compliance Targets

12.3.2.1 Accessibility Standards

NFR-ID	Standard	Target Level	v1 Status	v1.1 Status	Priority
NFR-ACC-001	WCAG 2.1	Level AA	Scaffold	Full Compliance	Must Have
NFR-ACC-002	Section 508	Full	Scaffold	Full Compliance	Must Have
NFR-ACC-003	WCAG 2.1	Level AAA	N/A	Partial	Nice to Have

12.3.2.2 Target Audience

NFR-ID	User Group	Accommodations	Priority
NFR-ACC-005	Visual impairments	Screen reader compatibility, high contrast	Must Have
NFR-ACC-006	Motor impairments	Full keyboard navigation, large touch targets	Must Have
NFR-ACC-007	Cognitive impairments	Clear language, consistent navigation	Should Have
NFR-ACC-008	Hearing impairments	Text alternatives for audio content	Should Have

12.3.3 Semantic HTML Requirements

12.3.3.1 Document Structure

NFR-ID	Requirement	Implementation	Verification	Priority
NFR-ACC-010	Semantic elements	Use <header>, <nav>, <main>, <footer>, <section>, <article>	Automated audit	Must Have
NFR-ACC-011	Heading hierarchy	<h1> to <h6> in logical order, no skips	Automated audit	Must Have

NFR-ACC-012	Landmark regions	All pages have identifiable main content area	Screen reader test	Must Have
NFR-ACC-013	Semantic lists	Use , for list content	Semantic validation	Must Have
NFR-ACC-014	Data tables	Use <table> with <th>, <caption> for tabular data	Table structure audit	Must Have

12.3.3.2 Page Layout Template

```
// Required page structure for all screens
<PageLayout>
  <header role="banner">...</header>
  <nav role="navigation">...</nav>
  <main role="main" id="main-content">...</main>
  <footer role="contentinfo">...</footer>
</PageLayout>
```

12.3.3.3 Verification Methods

- axe-core automated testing in CI/CD
- HTML validation via W3C validator
- Manual semantic structure review

12.3.4 Keyboard Navigation Requirements

12.3.4.1 Core Keyboard Requirements

NFR-ID	Requirement	Implementation	Verification	Priority
NFR-ACC-020	Tab order	Native focus order, avoid tabindex > 0	Manual tab test	Must Have
NFR-ACC-021	Focus visibility	Visible focus indicator (never outline: none alone)	Visual inspection	Must Have
NFR-ACC-022	Skip link	First focusable element: "Skip to main content"	Screen reader test	Must Have
NFR-ACC-023	Modal focus trap	Modals contain focus; Escape closes	Keyboard-only test	Must Have
NFR-ACC-024	No keyboard traps	All elements keyboard-escapable	Keyboard navigation test	Must Have
NFR-ACC-025	Arrow key navigation	Implemented for tabs, menus, listboxes	Component test	Should Have

12.3.4.2 Focus Indicator Standards

NFR-ID	Element Type	Focus Style	Priority
NFR-ACC-026	Interactive elements	2px ring with 2px offset	Must Have
NFR-ACC-027	Form inputs	Visible border change + ring	Must Have
NFR-ACC-028	Links	Underline + color change or ring	Must Have
NFR-ACC-029	Custom components	Equivalent visible focus	Must Have

12.3.4.3 Keyboard Shortcuts (v1.1)

NFR-ID	Shortcut	Action	Priority
NFR-ACC-030	/	Focus search field	Nice to Have
NFR-ACC-031	?	Open keyboard shortcuts help	Nice to Have
NFR-ACC-032	Escape	Close modals, cancel operations	Must Have

12.3.5 Form Accessibility Requirements

12.3.5.1 Form Field Requirements

NFR-ID	Requirement	Implementation	Priority
NFR-ACC-040	Label association	Every <input> has associated <label> via htmlFor	Must Have
NFR-ACC-041	Error identification	Errors linked via aria-describedby	Must Have
NFR-ACC-042	Required fields	aria-required="true" + visual indicator	Must Have
NFR-ACC-043	Error state	aria-invalid="true" when validation fails	Must Have
NFR-ACC-044	Input purpose	autocomplete attributes for user data	Should Have
NFR-ACC-045	Error prevention	Confirmation for destructive actions	Must Have

12.3.5.2 Form Pattern Template

```
<div>
  <label htmlFor="store-name" id="store-name-label">
    Store Name <span aria-hidden="true">*</span>
  </label>
  <input
    id="store-name"
    aria-labelledby="store-name-label"
    aria-describedby={error ? "store-name-error" : undefined}
    aria-required="true"
    aria-invalid={!error}
  />
  {error && (
    <span id="store-name-error" role="alert">
```

```
        {error}  
      </span>  
    )}  
</div>
```

12.3.5.3 Verification Methods

- Form association validation via axe-core
- Screen reader announcement testing
- Error message accessibility review

12.3.6 Color and Contrast Requirements

12.3.6.1 Contrast Ratio Requirements

NFR-ID	Element Type	Minimum Ratio	WCAG Criterion	Priority
NFR-ACC-050	Normal text (< 18pt)	4.5:1	1.4.3 AA	Must Have
NFR-ACC-051	Large text (>= 18pt or 14pt bold)	3:1	1.4.3 AA	Must Have
NFR-ACC-052	UI components (borders, icons)	3:1	1.4.11 AA	Must Have
NFR-ACC-053	Focus indicators	3:1	1.4.11 AA	Must Have
NFR-ACC-054	Enhanced contrast (AAA)	7:1	1.4.6 AAA	Nice to Have

12.3.6.2 Color Independence

NFR-ID	Requirement	Implementation	Priority
NFR-ACC-055	Color not sole indicator	Use icons, text, patterns in addition to color	Must Have
NFR-ACC-056	Status indicators	Icon + text + color for all statuses	Must Have
NFR-ACC-057	Error states	Red + icon + text description	Must Have
NFR-ACC-058	Links in text	Underline or non-color visual distinction	Must Have

12.3.6.3 Accessible Color Palette

Token	Value	Usage	Contrast on White
--text-primary	#1f2937 (gray-800)	Body text	12.6:1
--text-secondary	#4b5563 (gray-600)	Secondary text	7.0:1
--text-on-primary	#ffffff	Text on primary buttons	4.5:1 (on blue-600)
--error-text	#991b1b (red-800)	Error messages	9.4:1
--success-text	#166534 (green-800)	Success messages	7.1:1

12.3.6.4 Verification Methods

- Automated contrast checker in CI/CD
- Colorblind simulation testing
- Design review for color-only indicators

12.3.7 Interactive Component Patterns

12.3.7.1 ARIA Patterns by Component

NFR-ID	Component	ARIA Pattern	Keyboard Support	Priority
NFR-ACC-060	Buttons	<code><button></code> Or <code>role="button"</code> + <code>tabindex="0"</code>	Enter, Space	Must Have
NFR-ACC-061	Links	<code></code> with descriptive text	Enter	Must Have
NFR-ACC-062	Modals	<code>role="dialog"</code> + <code>aria-modal="true"</code> + <code>aria-labelledby</code>	Escape to close	Must Have
NFR-ACC-063	Tabs	<code>role="tablist/tab/tabpanel"</code>	Arrow keys	Must Have
NFR-ACC-064	Dropdowns	<code>role="listbox"</code> or native <code><select></code>	Arrow keys, Enter	Must Have
NFR-ACC-065	Alerts	<code>role="alert"</code> for important messages	Auto-announced	Must Have
NFR-ACC-066	Loading states	<code>aria-live="polite"</code> + <code>aria-busy</code>	Announce completion	Should Have
NFR-ACC-067	Progress bars	<code>role="progressbar"</code> + <code>aria-valuenow</code>	Screen reader value	Should Have

12.3.7.2 Prohibited Patterns

NFR-ID	Prohibited Pattern	Correct Alternative	Priority
NFR-ACC-068	<code><div></code> as button	<code><button></code> element	Must Have
NFR-ACC-069	"Click here" link text	Descriptive link text	Must Have
NFR-ACC-070	Icon-only buttons without labels	<code>aria-label</code> on button	Must Have
NFR-ACC-071	Auto-playing media	User-initiated playback	Must Have

12.3.8 Image and Media Requirements

12.3.8.1 Alternative Text

NFR-ID	Image Type	Alt Text Requirement	Priority
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NFR-ACC-080	Informative images	Descriptive alt text	Must Have
NFR-ACC-081	Decorative images	alt="" Or role="presentation"	Must Have
NFR-ACC-082	Complex images (charts)	aria-describedby linking to text description	Should Have
NFR-ACC-083	Icons in buttons	aria-label on parent button	Must Have

12.3.8.2 Media Alternatives

NFR-ID	Media Type	Alternative Required	Priority
NFR-ACC-085	Video content	Captions (v1.1)	Should Have
NFR-ACC-086	Audio content	Transcript (v1.1)	Should Have
NFR-ACC-087	Animated content	Pause/stop control	Should Have

12.3.9 Mobile/PWA Accessibility

12.3.9.1 Touch Target Requirements

NFR-ID	Requirement	Specification	Priority
NFR-ACC-090	Minimum touch target	44x44 CSS pixels	Must Have
NFR-ACC-091	Touch target spacing	8px minimum between targets	Should Have
NFR-ACC-092	Touch target padding	12px internal padding	Should Have

12.3.9.2 Responsive Accessibility

NFR-ID	Requirement	Specification	Priority
NFR-ACC-095	Text resize	Content readable at 200% zoom	Must Have
NFR-ACC-096	Orientation	Works in portrait and landscape	Must Have
NFR-ACC-097	Motion preferences	Respect prefers-reduced-motion	Must Have
NFR-ACC-098	Reflow	No horizontal scroll at 320px width	Should Have

12.3.9.3 Motion Preference Pattern

```
@media (prefers-reduced-motion: reduce) {  
  * {  
    animation-duration: 0.01ms !important;  
    animation-iteration-count: 1 !important;  
    transition-duration: 0.01ms !important;  
  }  
}
```

12.3.10 Screen Reader Support

12.3.10.1 v1 Screen Reader Compatibility

NFR-ID	Requirement	Implementation	Priority
NFR-ACC-100	Basic screen reader support	Semantic HTML, ARIA labels	Must Have
NFR-ACC-101	Form field announcements	Associated labels, error descriptions	Must Have
NFR-ACC-102	Dynamic content updates	aria-live regions for updates	Should Have
NFR-ACC-103	Navigation announcements	Page title updates, route changes	Should Have

12.3.10.2 v1.1 Screen Reader Testing Matrix

NFR-ID	Screen Reader	Platform	v1.1 Target	Priority
NFR-ACC-105	VoiceOver	iOS/macOS	Full testing pass	Should Have
NFR-ACC-106	NVDA	Windows	Full testing pass	Should Have
NFR-ACC-107	JAWS	Windows	Full testing pass	Nice to Have
NFR-ACC-108	TalkBack	Android	Full testing pass	Should Have

12.3.11 Testing and Validation

12.3.11.1 Automated Testing Requirements

NFR-ID	Tool	Integration Point	Scope	Priority
NFR-ACC-110	axe-core	CI/CD pipeline	All pages	Must Have
NFR-ACC-111	Lighthouse	Deployment check	Performance + accessibility	Must Have
NFR-ACC-112	Pa11y	Pre-release	Full application	Should Have

12.3.11.2 Manual Testing Requirements

NFR-ID	Test Type	Frequency	v1 Status	Priority
NFR-ACC-115	Keyboard-only navigation	Every sprint	Required	Must Have
NFR-ACC-116	Focus order verification	Every sprint	Required	Must Have
NFR-ACC-117	Color contrast audit	Monthly	Required	Must Have
NFR-ACC-118	200% zoom test	Monthly	Required	Must Have
NFR-ACC-119	Screen reader testing	Quarterly (v1.1)	Deferred	Should Have

12.3.11.3 CI/CD Gate

```
- name: Accessibility Check
run: |
  npx axe-core --tags wcag2a,wcag2aa
  # Fail on critical/serious issues
```

12.3.12 Documentation Requirements (v1.1)

12.3.12.1 Compliance Documentation

NFR-ID	Document	Purpose	v1.1 Target	Priority
NFR-ACC-120	VPAT	Voluntary Product Accessibility Template	Required	Should Have
NFR-ACC-121	Accessibility Statement	Public accessibility commitment	Required	Should Have
NFR-ACC-122	Keyboard shortcuts guide	User documentation	Recommended	Nice to Have
NFR-ACC-123	Screen reader guide	User documentation	Recommended	Nice to Have

12.3.13 Cross-References

Reference	Description
SUPP-038	Section 508 Accessibility Scaffold source document
SUPP-036	Screen and interface design patterns
WCAG 2.1 Guidelines	Full W3C specification
Section 508 Standards	Federal accessibility requirements

Document Status: Complete IEEE 830 Compliance: Section 3.3 - Accessibility Requirements

12.4 Scalability Requirements

IEEE 830 Reference: Section 3.3 - Non-Functional Requirements: Scalability **Source Documents:** SUPP-039 (Infrastructure, Performance & Security Requirements), SUPP-012 (Technology Selections) **Version:** 1.0 **Last Updated:** 2026-01-01

12.4.1 Purpose

This section defines scalability requirements for NewPOPSys v1, covering horizontal and vertical scaling strategies, load balancing, caching, and capacity planning. Requirements ensure the platform can grow from pilot scale to enterprise deployments.

12.4.2 Scaling Targets

12.4.2.1 Capacity Milestones

NFR-ID	Milestone	Store Count	Concurrent Users	Monthly Cost Target	Priority
NFR-SCL-001	Pilot	100 stores	10 users	\$500/month	Must Have
NFR-SCL-002	v1 Launch	500 stores	50 users	\$1,000/month	Must Have
NFR-SCL-003	v1.1 Growth	2,000 stores	200 users	\$2,500/month	Should Have
NFR-SCL-004	v2 Enterprise	10,000 stores	1,000 users	\$6,000/month	Nice to Have

12.4.2.2 Scaling Triggers

NFR-ID	Metric	Scale-Up Trigger	Scale-Down Trigger	Priority
NFR-SCL-005	CPU utilization	> 70% for 5 min	< 30% for 15 min	Must Have
NFR-SCL-006	Memory utilization	> 80% for 5 min	< 40% for 15 min	Must Have
NFR-SCL-007	Request latency (p95)	> 500ms for 5 min	N/A	Must Have
NFR-SCL-008	Queue depth	> 1,000 jobs	< 100 jobs	Should Have

12.4.3 Horizontal Scaling

12.4.3.1 Application Layer Scaling

NFR-ID	Component	Min Instances	Max Instances	Scaling Strategy	Priority
NFR-SCL-010	API containers	2	16	CPU/memory-based auto-scaling	Must Have
NFR-SCL-011	Worker containers	2	8	Queue depth-based scaling	Must Have
NFR-SCL-012	WebSocket servers	2	8	Connection count scaling	Should Have

12.4.3.2 Auto-Scaling Configuration

NFR-ID	Parameter	Value	Notes	Priority
NFR-SCL-015	Scale-up cooldown	60 seconds	Prevent thrashing	Must Have

NFR-SCL-016	Scale-down cooldown	300 seconds	Ensure stability	Must Have
NFR-SCL-017	Desired capacity buffer	+ 1 instance	Headroom for traffic spikes	Should Have
NFR-SCL-018	Health check interval	30 seconds	Detect unhealthy instances	Must Have

12.4.3.3 Stateless Design Requirements

NFR-ID	Requirement	Implementation	Priority
NFR-SCL-020	Session externalization	Redis-backed sessions	Must Have
NFR-SCL-021	No local file storage	S3 for all file operations	Must Have
NFR-SCL-022	Shared cache layer	Redis for application cache	Must Have
NFR-SCL-023	Idempotent operations	Request ID tracking	Must Have

12.4.4 Vertical Scaling

12.4.4.1 Database Vertical Scaling Path

NFR-ID	Instance Size	vCPU	Memory	Max Connections	Store Capacity	Priority
NFR-SCL-030	db.t3.medium	2	4 GB	100	100 stores	Must Have (Pilot)
NFR-SCL-031	db.r6g.large	2	16 GB	200	500 stores	Must Have (v1)
NFR-SCL-032	db.r6g.xlarge	4	32 GB	400	2,000 stores	Should Have
NFR-SCL-033	db.r6g.2xlarge	8	64 GB	800	10,000 stores	Nice to Have

12.4.4.2 Container Sizing

NFR-ID	Container Type	vCPU	Memory	Use Case	Priority
NFR-SCL-035	API (standard)	1	2 GB	Normal operations	Must Have
NFR-SCL-036	API (high-load)	2	4 GB	Campaign launch periods	Should Have
NFR-SCL-037	Worker (standard)	0.5	1 GB	Background jobs	Must Have
NFR-SCL-038	Worker (export)	2	4 GB	Large export jobs	Should Have

12.4.5 Database Scaling Strategy

12.4.5.1 Read Replica Configuration

NFR-ID	Requirement	Implementation	Activation Threshold	Priority
NFR-SCL-040	Read replica	RDS read replica in same region	> 500 stores	Should Have
NFR-SCL-041	Read replica routing	Application-level routing for read queries	> 500 stores	Should Have
NFR-SCL-042	Replica lag monitoring	Alert if lag > 10 seconds	Always	Must Have
NFR-SCL-043	Cross-region replica	DR region read replica	> 2,000 stores	Nice to Have

12.4.5.2 Connection Pooling

NFR-ID	Requirement	Value	Notes	Priority
NFR-SCL-045	Pool size per instance	20 connections	Based on (vCPU * 2) + 2	Must Have
NFR-SCL-046	Total pool limit	80% of max_connections	Reserve for admin	Must Have
NFR-SCL-047	Connection timeout	5 seconds	Fail fast	Must Have
NFR-SCL-048	Idle timeout	30 seconds	Release unused connections	Should Have

12.4.5.3 Query Optimization

NFR-ID	Requirement	Implementation	Priority
NFR-SCL-050	Index coverage	All filtered columns indexed	Must Have
NFR-SCL-051	Query planning	EXPLAIN ANALYZE for slow queries	Must Have
NFR-SCL-052	Materialized views	Dashboard aggregations	Should Have
NFR-SCL-053	Partition strategy	Time-based for audit logs (v1.1)	Nice to Have

12.4.6 Caching Strategy

12.4.6.1 Cache Layer Architecture

NFR-ID	Cache Layer	Technology	Scope	Priority
NFR-SCL-060	Browser cache	HTTP Cache-Control	Static assets	Must Have

NFR-SCL-061	CDN cache	CloudFront	Static assets, signed URLs	Must Have
NFR-SCL-062	Application cache	Redis	Session, reference data	Must Have
NFR-SCL-063	Query cache	Redis	Frequently accessed lists	Should Have
NFR-SCL-064	Database cache	PostgreSQL shared_buffers	Query results	Must Have

12.4.6.2 Cache Configuration

NFR-ID	Cache Type	TTL	Invalidation	Priority
NFR-SCL-065	Static assets (CDN)	1 year	Content-hash filenames	Must Have
NFR-SCL-066	Session data	15 min sliding	Logout/expiration	Must Have
NFR-SCL-067	Reference data	5 min	Background refresh	Should Have
NFR-SCL-068	List queries	1 min	Write-through invalidation	Should Have
NFR-SCL-069	User permissions	5 min	Role change event	Must Have

12.4.6.3 Redis Scaling

NFR-ID	Configuration	Value	Store Capacity	Priority
NFR-SCL-070	Single node	cache.t3.medium	Up to 500 stores	Must Have
NFR-SCL-071	Cluster mode	3 nodes	500-5,000 stores	Should Have
NFR-SCL-072	Memory allocation	2-4 GB	Based on session count	Must Have

12.4.7 Load Balancing

12.4.7.1 Load Balancer Configuration

NFR-ID	Requirement	Implementation	Priority
NFR-SCL-080	Application Load Balancer	AWS ALB	Must Have
NFR-SCL-081	Health check path	/health endpoint	Must Have
NFR-SCL-082	Health check interval	30 seconds	Must Have
NFR-SCL-083	Unhealthy threshold	2 consecutive failures	Must Have
NFR-SCL-084	Connection draining	30 seconds	Must Have

12.4.7.2 Routing Strategy

NFR-ID	Route Type	Distribution	Priority
NFR-SCL-085	HTTP/HTTPS traffic	Round-robin with health checks	Must Have
NFR-SCL-086	WebSocket connections	Sticky sessions (source IP)	Should Have
NFR-SCL-087	API rate limiting	Per-tenant, per-endpoint	Must Have

12.4.7.3 SSL/TLS Configuration

NFR-ID	Requirement	Specification	Priority
NFR-SCL-090	TLS version	1.3 minimum	Must Have
NFR-SCL-091	Certificate management	AWS ACM auto-renewal	Must Have
NFR-SCL-092	HTTPS redirect	Automatic HTTP to HTTPS	Must Have

12.4.8 Storage Scaling

12.4.8.1 Object Storage (S3)

NFR-ID	Requirement	Implementation	Priority
NFR-SCL-100	Bucket structure	Per-tenant prefix isolation	Must Have
NFR-SCL-101	Upload pattern	Direct-to-S3 presigned URLs	Must Have
NFR-SCL-102	Lifecycle policies	Transition to IA after 90 days	Should Have
NFR-SCL-103	Glacier archival	Campaign media after 1 year	Nice to Have

12.4.8.2 Storage Capacity Planning

NFR-ID	Store Count	Monthly Upload	Annual Storage	Priority
NFR-SCL-105	100 stores	10 GB	120 GB	Must Have
NFR-SCL-106	500 stores	50 GB	600 GB	Must Have
NFR-SCL-107	2,000 stores	200 GB	2.4 TB	Should Have
NFR-SCL-108	10,000 stores	1 TB	12 TB	Nice to Have

12.4.9 Background Job Scaling

12.4.9.1 Queue Architecture

NFR-ID	Queue	Purpose	Concurrency	Priority
NFR-SCL-110	default	Standard jobs (email, thumbnails)	10 per worker	Must Have
NFR-SCL-111	long	Exports, bulk operations	2 per worker	Must Have

NFR-SCL-112	critical	Webhooks, time-sensitive	20 per worker	Must Have
NFR-SCL-113	dead-letter	Failed job investigation	Manual processing	Must Have

12.4.9.2 Job Scaling Strategy

NFR-ID	Condition	Action	Priority
NFR-SCL-115	Queue depth > 1,000	Add worker instance	Should Have
NFR-SCL-116	Queue depth > 5,000	Alert + auto-scale	Must Have
NFR-SCL-117	Worker idle > 15 min	Remove worker instance	Should Have
NFR-SCL-118	Dead-letter > 50	Alert operations team	Must Have

12.4.10 CDN and Edge Caching

12.4.10.1 CloudFront Configuration

NFR-ID	Requirement	Implementation	Priority
NFR-SCL-120	Global distribution	CloudFront edge locations	Must Have
NFR-SCL-121	Origin shield	Enable for reduced origin load	Should Have
NFR-SCL-122	Cache behaviors	Static (long), API (short/no cache)	Must Have
NFR-SCL-123	Signed URLs	Photo access with expiration	Must Have

12.4.10.2 Cache Behavior Rules

NFR-ID	Path Pattern	Cache TTL	Signed	Priority
NFR-SCL-125	/_next/static/*	1 year	No	Must Have
NFR-SCL-126	/api/*	No cache	No	Must Have
NFR-SCL-127	/media/*	1 day	Yes	Must Have
NFR-SCL-128	/thumbnails/*	7 days	Yes	Must Have

12.4.11 Multi-Tenancy Scaling

12.4.11.1 Tenant Isolation

NFR-ID	Requirement	Implementation	Priority
NFR-SCL-130	Logical isolation	Tenant ID in all queries	Must Have
NFR-SCL-131	Resource quotas	Per-tenant limits (API, storage)	Should Have
NFR-SCL-132	Noisy neighbor prevention	Rate limiting per tenant	Must Have

NFR-SCL-133	Tenant monitoring	Per-tenant metrics dashboards	Should Have
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12.4.11.2 Tenant-Level Rate Limits

NFR-ID	Resource	Limit per Tenant	Burst	Priority
NFR-SCL-135	API requests	1,000/min	2,000/min	Must Have
NFR-SCL-136	Photo uploads	100/min	200/min	Must Have
NFR-SCL-137	Export jobs	10/hour	20/hour	Should Have
NFR-SCL-138	Webhook endpoints	5 per tenant	N/A	Must Have

12.4.12 Verification and Testing

12.4.12.1 Load Testing Requirements

NFR-ID	Test Type	Frequency	Target	Priority
NFR-SCL-140	Baseline load test	Each release	100 concurrent users	Must Have
NFR-SCL-141	Stress test	Monthly	2x expected peak	Should Have
NFR-SCL-142	Spike test	Quarterly	5x normal traffic burst	Should Have
NFR-SCL-143	Endurance test	Pre-launch	24-hour sustained load	Should Have

12.4.12.2 Scaling Verification

NFR-ID	Scenario	Expected Behavior	Priority
NFR-SCL-145	Traffic spike	Auto-scale within 2 minutes	Must Have
NFR-SCL-146	Traffic decrease	Scale down within 10 minutes	Should Have
NFR-SCL-147	Instance failure	Traffic rerouted within 30 seconds	Must Have
NFR-SCL-148	Database failover	Automatic failover within 2 minutes	Should Have

12.4.13 Cross-References

Reference	Description
SUPP-039	Infrastructure sizing and scaling projections
SUPP-012	Technology stack selections
Section 3.3	Technology Stack documentation
Section 12.1	Performance requirements
Section 12.5	Availability requirements

12.5 Availability Requirements

IEEE 830 Reference: Section 3.3 - Non-Functional Requirements: Availability **Source Documents:** SUPP-039 (Infrastructure, Performance & Security Requirements) **Version:** 1.0 **Last Updated:** 2026-01-01

12.5.1 Purpose

This section defines availability requirements for NewPOPSys v1, covering uptime SLAs, disaster recovery, failover mechanisms, backup strategies, and maintenance procedures. All requirements include specific targets and verification methods.

12.5.2 Uptime Requirements

12.5.2.1 Service Level Targets

NFR-ID	Environment	Uptime Target	Allowed Downtime/Month	Priority
NFR-AVL-001	Production	99.5%	3.6 hours	Must Have
NFR-AVL-002	Staging	95%	36 hours	Should Have
NFR-AVL-003	Development	90%	72 hours	Nice to Have

12.5.2.2 Uptime Calculation

NFR-ID	Requirement	Specification	Priority
NFR-AVL-005	Measurement period	Calendar month	Must Have
NFR-AVL-006	Measurement method	External synthetic monitoring	Must Have
NFR-AVL-007	Exclusions	Scheduled maintenance windows	Must Have
NFR-AVL-008	Reporting	Monthly availability report	Should Have

12.5.2.3 Availability by Component

NFR-ID	Component	Target	Dependency	Priority
NFR-AVL-010	Web application	99.5%	ALB, API containers	Must Have
NFR-AVL-011	API endpoints	99.5%	ALB, API containers, Database	Must Have
NFR-AVL-012	Database	99.95%	RDS Multi-AZ	Must Have
NFR-AVL-013	Object storage	99.99%	S3 (AWS SLA)	Must Have
NFR-AVL-014	CDN	99.9%	CloudFront	Must Have
NFR-AVL-015	Background jobs	99%	Redis, Worker containers	Should Have

12.5.3 High Availability Architecture

12.5.3.1 Multi-AZ Deployment

NFR-ID	Component	AZ Configuration	Failover	Priority
NFR-AVL-020	Application containers	2+ AZs	Automatic via ALB	Must Have
NFR-AVL-021	Database (RDS)	Multi-AZ standby	Automatic failover	Must Have
NFR-AVL-022	Redis (ElastiCache)	Multi-AZ replication	Automatic failover	Should Have
NFR-AVL-023	NAT Gateway	Per-AZ deployment	No single point of failure	Should Have

12.5.3.2 Load Balancer Configuration

NFR-ID	Requirement	Specification	Priority
NFR-AVL-025	Health check interval	30 seconds	Must Have
NFR-AVL-026	Healthy threshold	2 consecutive successes	Must Have
NFR-AVL-027	Unhealthy threshold	2 consecutive failures	Must Have
NFR-AVL-028	Health check path	/health (includes DB check)	Must Have
NFR-AVL-029	Connection draining	30 seconds	Must Have

12.5.3.3 Container Health Checks

NFR-ID	Check Type	Interval	Timeout	Retries	Priority
NFR-AVL-030	Startup probe	5s	10s	3	Must Have
NFR-AVL-031	Liveness probe	30s	5s	3	Must Have
NFR-AVL-032	Readiness probe	10s	5s	1	Must Have

12.5.4 Disaster Recovery

12.5.4.1 Recovery Objectives

NFR-ID	Scenario	RTO (Recovery Time)	RPO (Recovery Point)	Priority
NFR-AVL-040	Single AZ failure	< 5 minutes	0 (no data loss)	Must Have
NFR-AVL-041	Database failure	< 5 minutes	0 (Multi-AZ sync)	Must Have
NFR-AVL-042	Region failure	< 4 hours	< 1 hour	Should Have
NFR-AVL-043	Data corruption	< 1 hour	< 5 minutes (PITR)	Must Have
NFR-AVL-044	Ransomware/breach	< 4 hours	< 1 hour	Should Have

NFR-AVL-045	Complete rebuild	< 24 hours	< 1 hour	Should Have
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12.5.4.2 Disaster Recovery Strategy

NFR-ID	Tier	Strategy	Implementation	Priority
NFR-AVL-050	Tier 1 (v1)	Backup and restore	Daily backups to DR region	Must Have
NFR-AVL-051	Tier 2 (v1.1)	Pilot light	Cross-region DB replica (cold)	Should Have
NFR-AVL-052	Tier 3 (v2)	Warm standby	Active-passive cross-region	Nice to Have

12.5.4.3 Cross-Region Replication (v1.1+)

NFR-ID	Component	Replication Method	Lag Target	Priority
NFR-AVL-055	Database	RDS cross-region read replica	< 1 minute	Should Have
NFR-AVL-056	Object storage	S3 cross-region replication	< 15 minutes	Should Have
NFR-AVL-057	Configuration	Git-versioned + Secrets Manager	Immediate	Must Have

12.5.5 Backup Strategy

12.5.5.1 Database Backup

NFR-ID	Backup Type	Frequency	Retention	Priority
NFR-AVL-060	Automated daily snapshots	Daily	30 days	Must Have
NFR-AVL-061	Continuous WAL archiving	Continuous	7 days	Must Have
NFR-AVL-062	Weekly full backup	Weekly	90 days	Should Have
NFR-AVL-063	Monthly archival	Monthly	1 year	Nice to Have

12.5.5.2 Point-in-Time Recovery (PITR)

NFR-ID	Requirement	Specification	Priority
NFR-AVL-065	PITR window	7 days	Must Have
NFR-AVL-066	Recovery granularity	5-minute intervals	Must Have
NFR-AVL-067	Recovery testing	Quarterly DR drill	Should Have

12.5.5.3 Object Storage Backup

NFR-ID	Strategy	Implementation	Retention	Priority
NFR-AVL-070	Versioning	S3 versioning enabled	90 days	Must Have
NFR-AVL-071	Cross-region copy	S3 replication to DR region	Same as source	Should Have

NFR-AVL-072	Lifecycle rules	Transition to Glacier after 1 year	7 years	Nice to Have
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12.5.5.4 Configuration Backup

NFR-ID	Configuration Type	Backup Method	Priority
NFR-AVL-075	Infrastructure code	Git (GitHub/GitLab)	Must Have
NFR-AVL-076	Application config	Git + environment variables	Must Have
NFR-AVL-077	Secrets	AWS Secrets Manager (versioned)	Must Have
NFR-AVL-078	Database schema	Drizzle migrations in Git	Must Have

12.5.6 Failover Mechanisms

12.5.6.1 Automatic Failover

NFR-ID	Component	Failover Mechanism	Detection Time	Recovery Time	Priority
NFR-AVL-080	Application instances	ALB health checks	60 seconds	Immediate	Must Have
NFR-AVL-081	Database (Multi-AZ)	RDS automatic failover	60-120 seconds	60-120 seconds	Must Have
NFR-AVL-082	Redis (Multi-AZ)	ElastiCache failover	60 seconds	60 seconds	Should Have
NFR-AVL-083	DNS failover	Route 53 health checks	30 seconds	60 seconds	Should Have

12.5.6.2 Manual Failover Procedures

NFR-ID	Scenario	Procedure	Authorization Required	Priority
NFR-AVL-085	Promote read replica	Database failover runbook	Database admin	Should Have
NFR-AVL-086	Region switchover	DR activation runbook	Platform admin + approval	Should Have
NFR-AVL-087	Rollback deployment	Deployment rollback runbook	DevOps team	Must Have

12.5.6.3 Verification Methods

- Quarterly failover drills (documented)
- Annual DR simulation
- Chaos engineering tests (v1.1+)

12.5.7 Maintenance Windows

12.5.7.1 Scheduled Maintenance

NFR-ID	Maintenance Type	Window	Frequency	Notification	Priority
NFR-AVL-090	Application deployment	Any (zero-downtime)	As needed	Optional	Must Have
NFR-AVL-091	Database patching	Sunday 02:00-06:00 UTC	Monthly	7 days advance	Must Have
NFR-AVL-092	Infrastructure updates	Sunday 02:00-06:00 UTC	Quarterly	14 days advance	Should Have
NFR-AVL-093	Major version upgrades	Coordinated	As needed	30 days advance	Should Have

12.5.7.2 Zero-Downtime Deployment

NFR-ID	Requirement	Implementation	Priority
NFR-AVL-095	Rolling deployments	ECS rolling update	Must Have
NFR-AVL-096	Database migrations	Backward-compatible only	Must Have
NFR-AVL-097	Blue-green capability	For major changes	Should Have
NFR-AVL-098	Canary releases	Gradual traffic shift	Nice to Have

12.5.7.3 Emergency Maintenance

NFR-ID	Requirement	SLA	Authorization	Priority
NFR-AVL-100	Security patching	Within 24 hours of critical CVE	Security lead	Must Have
NFR-AVL-101	Critical bug fix	Within 4 hours of P1 incident	On-call engineer	Must Have
NFR-AVL-102	Emergency notification	30 minutes before if possible	On-call engineer	Must Have

12.5.8 Incident Response

12.5.8.1 Incident Severity Classification

NFR-ID	Severity	Description	Response Time	Resolution Target	Priority
NFR-AVL-110	P1 - Critical	Complete outage, data loss risk	15 minutes	1 hour	Must Have

NFR-AVL-111	P2 - High	Major feature unavailable	30 minutes	4 hours	Must Have
NFR-AVL-112	P3 - Medium	Degraded performance	2 hours	24 hours	Should Have
NFR-AVL-113	P4 - Low	Minor issue, workaround available	8 hours	7 days	Should Have

12.5.8.2 On-Call Requirements

NFR-ID	Requirement	Specification	Priority
NFR-AVL-115	On-call coverage	24/7/365	Must Have
NFR-AVL-116	Escalation path	Primary -> Secondary -> Manager	Must Have
NFR-AVL-117	Response acknowledgment	Within 15 minutes	Must Have
NFR-AVL-118	Incident documentation	Post-incident report within 48 hours	Should Have

12.5.8.3 Runbook Requirements

NFR-ID	Runbook	Purpose	Priority
NFR-AVL-120	Service restart	Container/service restart procedures	Must Have
NFR-AVL-121	Database recovery	PITR and failover procedures	Must Have
NFR-AVL-122	Deployment rollback	Quick rollback to previous version	Must Have
NFR-AVL-123	DR activation	Region failover procedures	Should Have
NFR-AVL-124	Security incident	Breach response procedures	Must Have

12.5.9 Monitoring and Alerting

12.5.9.1 Availability Monitoring

NFR-ID	Monitor Type	Check Interval	Threshold	Priority
NFR-AVL-130	Synthetic monitoring	1 minute	Response > 5s = warning	Must Have
NFR-AVL-131	Endpoint health checks	30 seconds	2 failures = alert	Must Have
NFR-AVL-132	Database connectivity	30 seconds	1 failure = alert	Must Have
NFR-AVL-133	SSL certificate expiry	Daily	< 30 days = warning	Must Have

12.5.9.2 Alert Channels

NFR-ID	Severity	Alert Channel	Priority
NFR-AVL-135	P1 Critical	PagerDuty + Phone call	Must Have

NFR-AVL-136	P2 High	PagerDuty + Slack	Must Have
NFR-AVL-137	P3 Medium	Slack + Email	Should Have
NFR-AVL-138	P4 Low	Email + Dashboard	Should Have

12.5.9.3 Status Communication

NFR-ID	Requirement	Implementation	Priority
NFR-AVL-140	Status page	Public status page (e.g., Statuspage.io)	Should Have
NFR-AVL-141	Incident updates	Every 30 minutes during P1/P2	Must Have
NFR-AVL-142	Maintenance announcements	In-app banner + email	Should Have

12.5.10 Graceful Degradation

12.5.10.1 Degradation Strategies

NFR-ID	Failure Scenario	Degraded Behavior	Priority
NFR-AVL-150	Redis unavailable	Fall back to database sessions	Should Have
NFR-AVL-151	S3 unavailable	Queue uploads, display placeholders	Should Have
NFR-AVL-152	Email service down	Queue emails, retry later	Must Have
NFR-AVL-153	Read replica unavailable	Route reads to primary	Should Have
NFR-AVL-154	CDN unavailable	Serve from origin directly	Should Have

12.5.10.2 Circuit Breaker Patterns

NFR-ID	Requirement	Specification	Priority
NFR-AVL-155	Failure threshold	50% failures in 10-second window	Should Have
NFR-AVL-156	Open circuit duration	30 seconds	Should Have
NFR-AVL-157	Half-open testing	3 successful requests to close	Should Have

12.5.11 Cross-References

Reference	Description
SUPP-039	Infrastructure availability and DR requirements
Section 12.1	Performance requirements
Section 12.4	Scalability requirements
Section 12.6	Data retention and backup policies

12.6 Data Retention Requirements

IEEE 830 Reference: Section 3.3 - Non-Functional Requirements: Data Retention **Source Documents:** SUPP-008 (Data Retention Classification Policy), SUPP-039 (Infrastructure Requirements) **Version:** 1.0 **Last Updated:** 2026-01-01

12.6.1 Purpose

This section defines data retention requirements for NewPOPSys v1, covering data classification, retention periods, purge workflows, GDPR compliance, backup policies, and audit trail retention. The core principle is: **Delete heavy artifacts on a clock; keep lightweight accountability.**

12.6.2 Data Classification

12.6.2.1 Data Classes Overview

NFR-ID	Class	Description	Examples	Retention Principle
NFR-RET-001	Class A: Heavy Media	High-storage campaign artifacts	Install photos, receipt photos, deinstall photos, optional videos	Time-limited (90 days default)
NFR-RET-002	Class B: Light Records	Campaign metadata and audit trail	Campaign records, assignments, statuses, review outcomes	Long horizon (2-5 years)
NFR-RET-003	Class C: Master Data	Core operational data	Stores, regions, groups, store layouts	While active
NFR-RET-004	Class D: Reusable Definitions	Templates and configurations	Survey templates, kit definitions, photo rule presets	While referenced
NFR-RET-005	Class E: Asset Registry	Metadata and pointers	Asset metadata, checksums, versions, pointer URIs	While referenced
NFR-RET-006	Class F: Exports	Generated output files	CSV, XLSX, JSON, PDF exports	Short-lived (regenerate on demand)

12.6.2.2 Storage Classification

NFR-ID	Class	Storage Type	Estimated Size per Campaign	Priority
NFR-RET-010	Class A	S3 Standard -> IA	500 MB - 5 GB	Must Have

NFR-RET-011	Class B	PostgreSQL	1-10 MB	Must Have
NFR-RET-012	Class C	PostgreSQL	< 1 MB	Must Have
NFR-RET-013	Class D	PostgreSQL	< 500 KB	Must Have
NFR-RET-014	Class E	PostgreSQL	< 100 KB	Must Have
NFR-RET-015	Class F	S3 (temporary)	Variable	Should Have

12.6.3 Retention Periods

12.6.3.1 Class A: Campaign Proof Media

NFR-ID	Requirement	Default	Configurable Range	Priority
NFR-RET-020	Retention after campaign completion	90 days	30-365 days	Must Have
NFR-RET-021	Export bundle notification	14 days before purge	7-30 days	Must Have
NFR-RET-022	Purge execution	Automated background job	N/A	Must Have
NFR-RET-023	Purge audit logging	Required	N/A	Must Have

12.6.3.2 Class B: Campaign Records (Light)

NFR-ID	Requirement	Recommended	Configurable	Priority
NFR-RET-025	Campaign metadata	5 years	Yes (2-indefinite)	Must Have
NFR-RET-026	Assignment records	5 years	Yes	Must Have
NFR-RET-027	Status history	5 years	Yes	Must Have
NFR-RET-028	Review outcomes	5 years	Yes	Must Have
NFR-RET-029	Audit events	7 years (compliance)	Minimum only	Must Have

12.6.3.3 Class C: Store Master Data

NFR-ID	Requirement	Retention	Notes	Priority
NFR-RET-030	Active stores	Indefinite	While store is active	Must Have
NFR-RET-031	Deactivated stores	2 years	After deactivation	Should Have
NFR-RET-032	Store layouts/surveys	Indefinite	While store exists	Must Have
NFR-RET-033	Regions and groups	Indefinite	While referenced	Must Have

12.6.3.4 Class D: Reusable Definitions

NFR-ID	Requirement	Retention	Notes	Priority
NFR-RET-035	Survey templates	While referenced	Plus 2 years after last use	Must Have
NFR-RET-036	Template versions	Indefinite	Immutable, never delete used versions	Must Have
NFR-RET-037	Kit/item definitions	While referenced	Plus 2 years after last use	Must Have
NFR-RET-038	Photo rule presets	While referenced	Supports repeat campaigns	Should Have

12.6.3.5 Class E: Asset Registry

NFR-ID	Requirement	Retention	Notes	Priority
NFR-RET-040	Asset metadata records	While binary exists	Pointer URI, checksum, version	Must Have
NFR-RET-041	Orphaned asset records	30 days	After binary purge	Should Have
NFR-RET-042	Asset tags and versions	With parent record	N/A	Must Have

12.6.3.6 Class F: Exports

NFR-ID	Requirement	Retention	Notes	Priority
NFR-RET-045	Generated exports	7 days	Regenerate on demand	Must Have
NFR-RET-046	Export request logs	90 days	Audit trail	Should Have
NFR-RET-047	Large exports	30 days	> 100 MB exports	Should Have

12.6.4 Purge Workflow

12.6.4.1 Automated Purge Process

NFR-ID	Step	Description	Timing	Priority
NFR-RET-050	Campaign completion trigger	Mark campaign as complete, schedule purge	Immediate	Must Have
NFR-RET-051	Export bundle generation	Generate downloadable export bundle	T-14 days	Must Have
NFR-RET-052	Purge notification	Notify tenant admins of pending purge	T-14 days	Must Have

NFR-RET-053	Reminder notification	Second notification before purge	T-3 days	Should Have
NFR-RET-054	Purge execution	Delete Class A media from S3	T+0	Must Have
NFR-RET-055	Metadata update	Mark media records as purged, retain manifest	T+0	Must Have
NFR-RET-056	Audit log	Log purge action with actor=system	T+0	Must Have

12.6.4.2 Purge Job Configuration

NFR-ID	Parameter	Value	Notes	Priority
NFR-RET-060	Job schedule	Daily at 02:00 UTC	Off-peak processing	Must Have
NFR-RET-061	Batch size	100 campaigns per run	Prevent resource exhaustion	Must Have
NFR-RET-062	Retry policy	3 retries with exponential backoff	Handle transient failures	Must Have
NFR-RET-063	Dead-letter handling	Alert on repeated failures	Manual investigation	Must Have

12.6.4.3 Export Bundle Contents

NFR-ID	Content	Format	Included	Priority
NFR-RET-065	Campaign summary	JSON + PDF	Always	Must Have
NFR-RET-066	Store completion status	CSV	Always	Must Have
NFR-RET-067	Verification outcomes	CSV	Always	Must Have
NFR-RET-068	Issues and reorders	CSV	Always	Must Have
NFR-RET-069	Audit events	JSON	Always	Must Have
NFR-RET-070	Media manifest	JSON	Always (IDs + pointers)	Must Have
NFR-RET-071	Media files (ZIP)	Binary	If within retention	Should Have

12.6.5 GDPR Compliance

12.6.5.1 Data Subject Rights

NFR-ID	Right	Implementation	Response Time	Priority
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NFR-RET-080	Right to access (Article 15)	Data export functionality	30 days	Must Have
NFR-RET-081	Right to rectification (Article 16)	Self-service profile edit + admin tools	Immediate	Must Have
NFR-RET-082	Right to erasure (Article 17)	Account deletion workflow	30 days	Must Have
NFR-RET-083	Right to portability (Article 20)	Standard export formats (JSON, CSV)	30 days	Should Have
NFR-RET-084	Right to object (Article 21)	Opt-out of marketing communications	Immediate	Should Have

12.6.5.2 Right to Erasure (RTBF) Workflow

NFR-ID	Step	Description	Notes	Priority
NFR-RET-090	Request submission	User or admin initiates deletion request	Via UI or support	Must Have
NFR-RET-091	Verification	Verify identity of requester	Email confirmation	Must Have
NFR-RET-092	Approval (if needed)	Brand Admin approval for store users	Optional workflow	Should Have
NFR-RET-093	Data collection	Identify all data associated with subject	Automated scan	Must Have
NFR-RET-094	Backup exclusion	Mark data for exclusion from future restores	Backup metadata	Should Have
NFR-RET-095	Data deletion	Remove/anonymize personal data	Preserve audit integrity	Must Have
NFR-RET-096	Confirmation	Notify requester of completion	Email notification	Must Have

12.6.5.3 Data Anonymization

NFR-ID	Data Type	Anonymization Method	Priority
NFR-RET-100	User names	Replace with "Deleted User [hash]"	Must Have
NFR-RET-101	Email addresses	Hash or remove	Must Have
NFR-RET-102	Phone numbers	Remove	Must Have
NFR-RET-103	IP addresses	Truncate to /24 or remove	Should Have
NFR-RET-104	User-generated content	Anonymize attribution	Should Have

12.6.5.4 GDPR Exceptions

NFR-ID	Exception	Justification	Retention	Priority
NFR-RET-105	Legal hold	Legal proceedings	Until released	Must Have
NFR-RET-106	Regulatory compliance	Audit trail requirements	Per regulation	Must Have
NFR-RET-107	Contractual obligation	Active contract terms	Contract duration	Should Have

12.6.6 Audit Trail Retention

12.6.6.1 Audit Log Types

NFR-ID	Audit Type	Retention	Purge Allowed	Priority
NFR-RET-110	Authentication events	2 years	After retention	Must Have
NFR-RET-111	Authorization events	2 years	After retention	Must Have
NFR-RET-112	Data access events	5 years	After retention	Must Have
NFR-RET-113	Data modification events	7 years	After retention	Must Have
NFR-RET-114	Administrative actions	7 years	After retention	Must Have
NFR-RET-115	System events (purge, sync)	2 years	After retention	Should Have

12.6.6.2 Audit Log Protection

NFR-ID	Requirement	Implementation	Priority
NFR-RET-120	Immutability	Append-only storage	Must Have
NFR-RET-121	Integrity verification	Checksums on log entries	Should Have
NFR-RET-122	Separate access control	Audit logs isolated from app permissions	Must Have
NFR-RET-123	Encryption at rest	AES-256 encryption	Must Have

12.6.6.3 Audit Log Archival

NFR-ID	Archive Stage	Timing	Storage	Priority
NFR-RET-125	Hot storage	0-90 days	PostgreSQL	Must Have
NFR-RET-126	Warm storage	90 days - 1 year	S3 Standard	Should Have
NFR-RET-127	Cold storage	1-7 years	S3 Glacier	Nice to Have

12.6.7 Backup Retention

12.6.7.1 Database Backup Retention

NFR-ID	Backup Type	Retention	Storage	Priority
NFR-RET-130	Daily snapshots	30 days	RDS snapshots	Must Have
NFR-RET-131	Weekly snapshots	90 days	RDS snapshots	Should Have
NFR-RET-132	Monthly snapshots	1 year	S3 (cross-region)	Nice to Have
NFR-RET-133	WAL archives (PITR)	7 days	RDS continuous	Must Have

12.6.7.2 Object Storage Retention

NFR-ID	Strategy	Configuration	Retention	Priority
NFR-RET-135	Versioning	Enabled on all buckets	90 days	Must Have
NFR-RET-136	Delete markers	Retained for recovery	30 days	Should Have
NFR-RET-137	Cross-region replication	DR region copy	Same as source	Should Have

12.6.7.3 Configuration Backup

NFR-ID	Configuration	Backup Method	Retention	Priority
NFR-RET-140	Infrastructure code	Git version control	Indefinite	Must Have
NFR-RET-141	Application config	Git + environment backup	Indefinite	Must Have
NFR-RET-142	Secrets	Secrets Manager versioning	30 versions	Must Have
NFR-RET-143	Database migrations	Git (Drizzle migrations)	Indefinite	Must Have

12.6.8 Tenant Data Lifecycle

12.6.8.1 Tenant Offboarding

NFR-ID	Step	Description	Timing	Priority
NFR-RET-150	Offboarding notice	Tenant provides termination notice	T-30 days	Must Have
NFR-RET-151	Export provision	Generate complete data export	T-14 days	Must Have
NFR-RET-152	Export download	Tenant downloads all data	T-7 days	Must Have
NFR-RET-153	Account deactivation	Disable tenant access	T+0	Must Have
NFR-RET-154	Data deletion	Delete tenant data (except audit)	T+30 days	Must Have
NFR-RET-155	Final confirmation	Confirm deletion to tenant	T+31 days	Must Have

12.6.8.2 Tenant Data Export (Takeout)

NFR-ID	Export Content	Format	Priority
NFR-RET-160	Store data	CSV + JSON	Must Have

NFR-RET-161	Campaign data	CSV + JSON	Must Have
NFR-RET-162	User data	JSON	Must Have
NFR-RET-163	Media files	ZIP archive	Must Have
NFR-RET-164	Configuration	JSON	Should Have
NFR-RET-165	Audit logs	JSON	Should Have

12.6.9 Storage Lifecycle Management

12.6.9.1 S3 Lifecycle Policies

NFR-ID	Object Type	Transition	Archive	Delete	Priority
NFR-RET-170	Campaign media	Standard -> IA (90d)	Glacier (1y)	After purge	Must Have
NFR-RET-171	Thumbnails	Standard only	N/A	With parent	Must Have
NFR-RET-172	Exports	Standard only	N/A	7 days	Must Have
NFR-RET-173	Audit log archives	Standard -> IA (90d)	Glacier (1y)	7 years	Should Have

12.6.9.2 Database Maintenance

NFR-ID	Maintenance	Frequency	Notes	Priority
NFR-RET-175	Vacuum analyze	Daily	Reclaim space	Must Have
NFR-RET-176	Index maintenance	Weekly	Rebuild fragmented indexes	Should Have
NFR-RET-177	Partition maintenance	Monthly (v1.1)	Audit log partitions	Nice to Have

12.6.10 Verification and Compliance

12.6.10.1 Retention Verification

NFR-ID	Verification	Frequency	Method	Priority
NFR-RET-180	Purge job execution	Daily	Job monitoring dashboard	Must Have
NFR-RET-181	Backup integrity	Weekly	Automated restore test	Should Have
NFR-RET-182	RTBF compliance	Per request	Verification checklist	Must Have
NFR-RET-183	Retention policy audit	Quarterly	Manual review	Should Have

12.6.10.2 Compliance Reporting

NFR-ID	Report	Frequency	Audience	Priority
NFR-RET-185	Data retention summary	Monthly	Internal ops	Should Have

NFR-RET-186	GDPR compliance report	Quarterly	DPO / Legal	Should Have
NFR-RET-187	Storage utilization	Monthly	Finance / Ops	Should Have

12.6.11 Cross-References

Reference	Description
SUPP-008	Data Retention Classification Policy (Option A)
SUPP-039	Infrastructure and backup requirements
SUPP-032	Break-glass, retention safety, and data requests
Section 12.2	Security requirements (audit logging)
Section 12.5	Availability requirements (backup strategy)

Document Status: Complete IEEE 830 Compliance: Section 3.3 - Data Retention Requirements